L2: Markov Decision Processes

EECE 576N | From MDPs to Deep RL | Fall 2026

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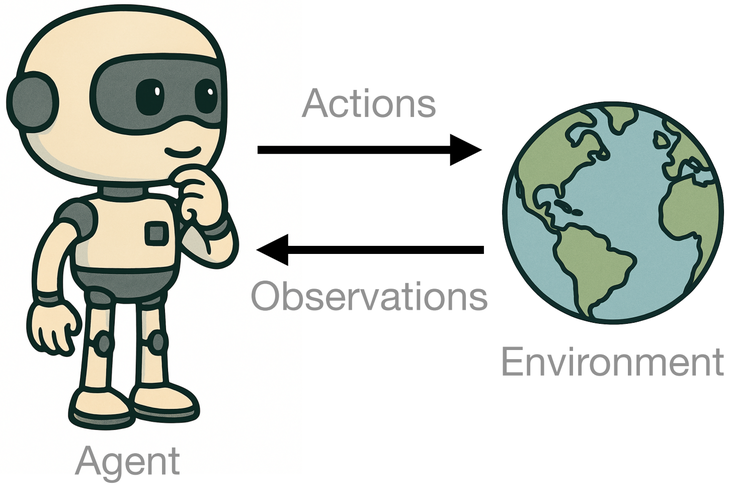
Learning Objectives

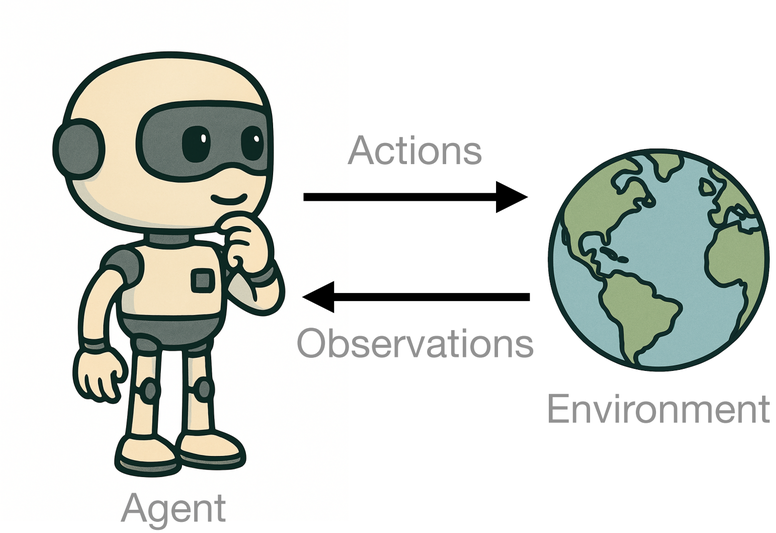
By the end of today's lecture, you should be able to…

- Explain the core modeling components of Markov decision processes (MDPs).
- Differentiate between classes of decision-making policies.
- Understand the components of the typical optimization objective in MDP problems.
- Define value functions and optimal value functions, and explain their role in sequential decision-making problems.



Sequential Decision-Making Problems

Objective: The decision-making *agent* should implement a *policy* that selects actions a_t at every *decision epoch* t to maximize some function of the resulting rewards $r_0, r_1, \ldots$



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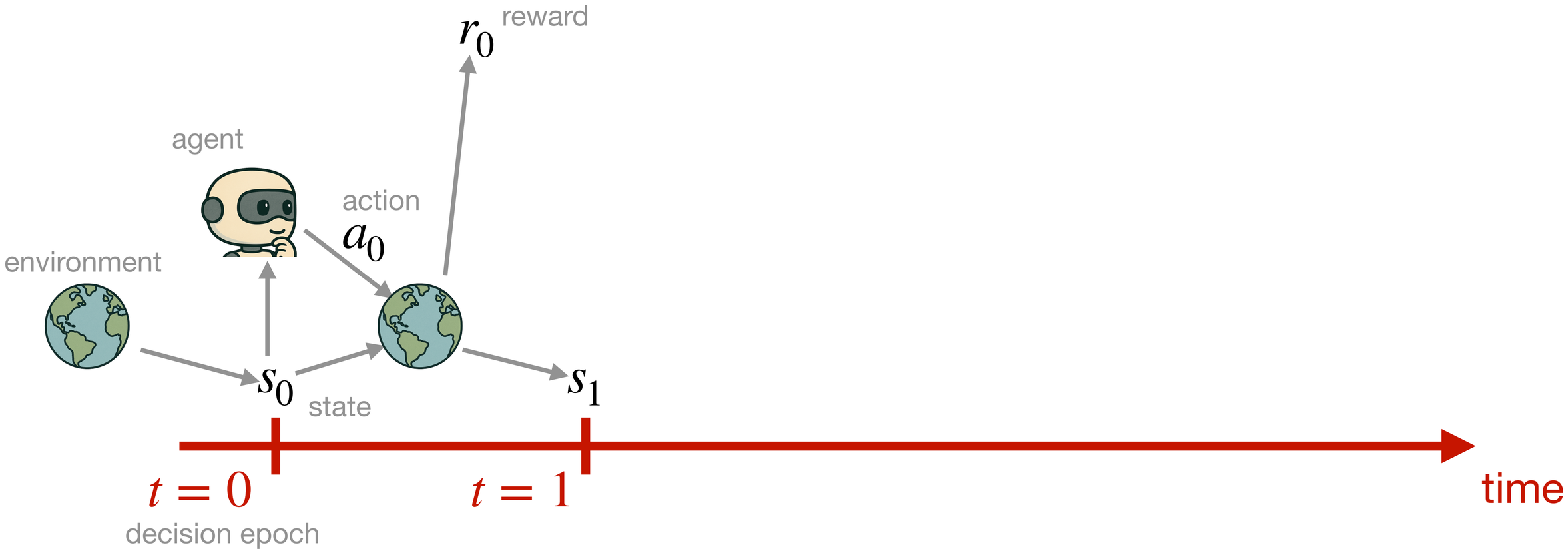

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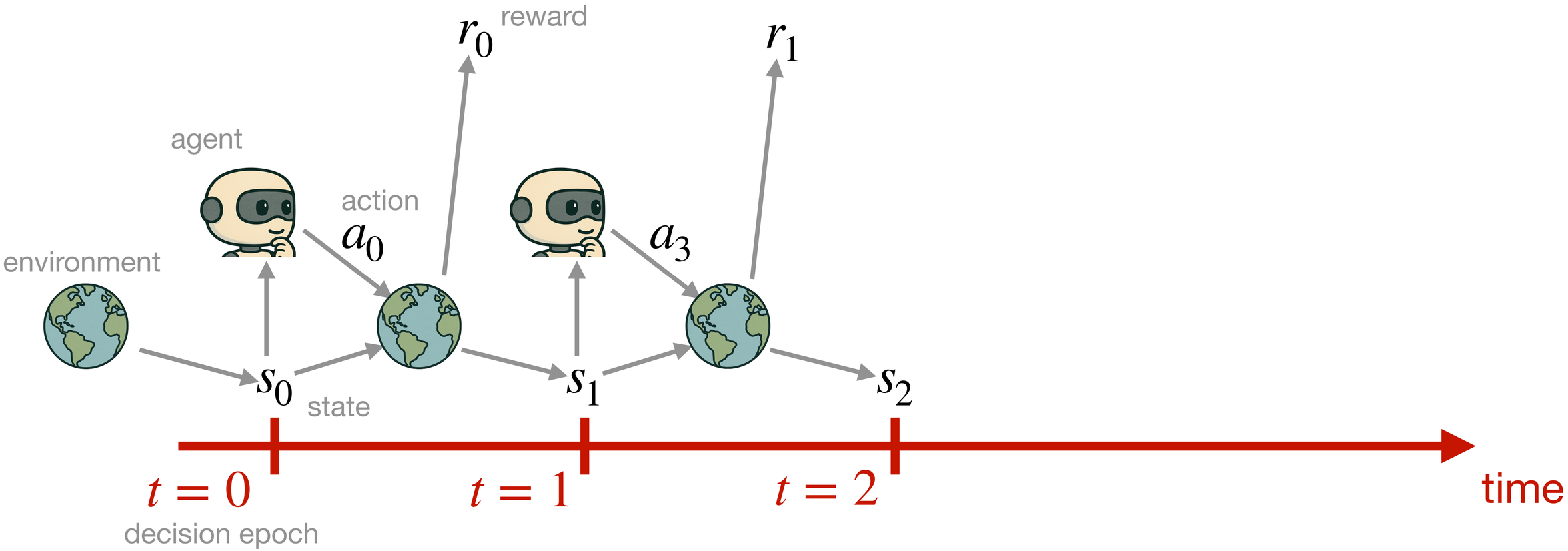

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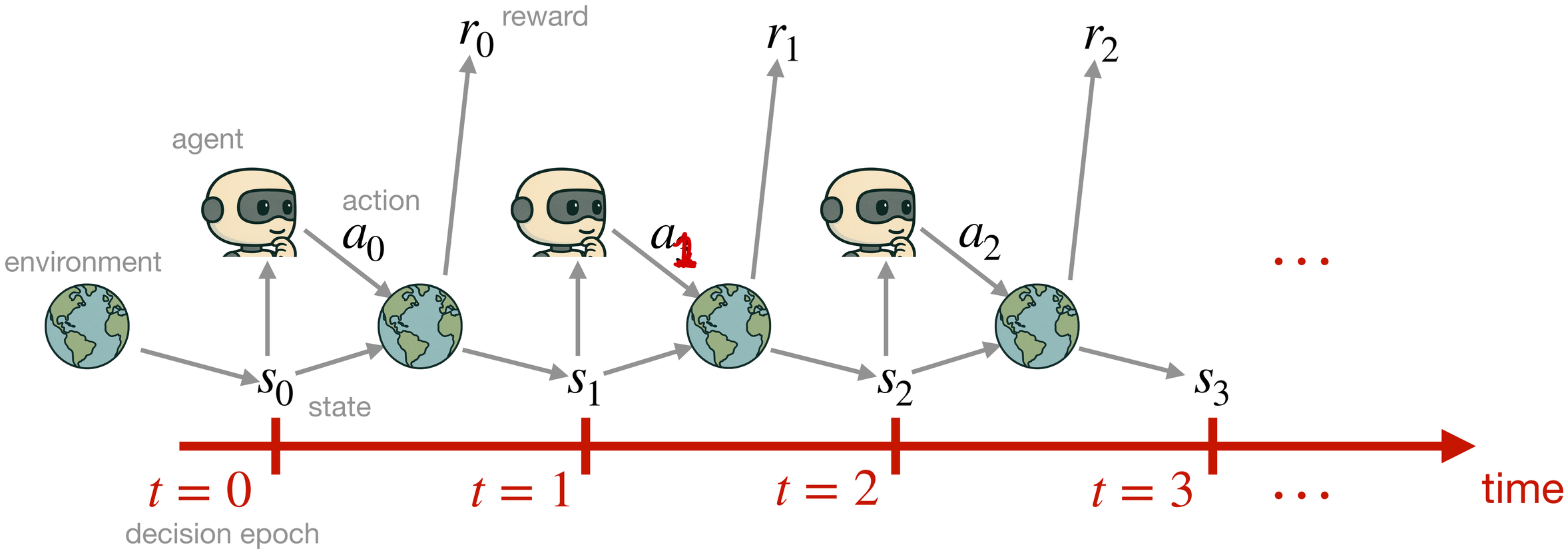
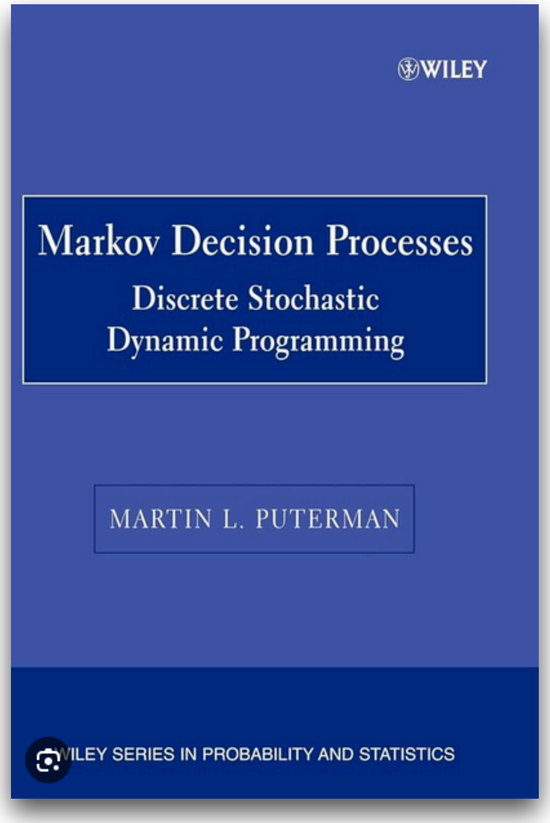


Modeling Sequential Decision-Making Problems



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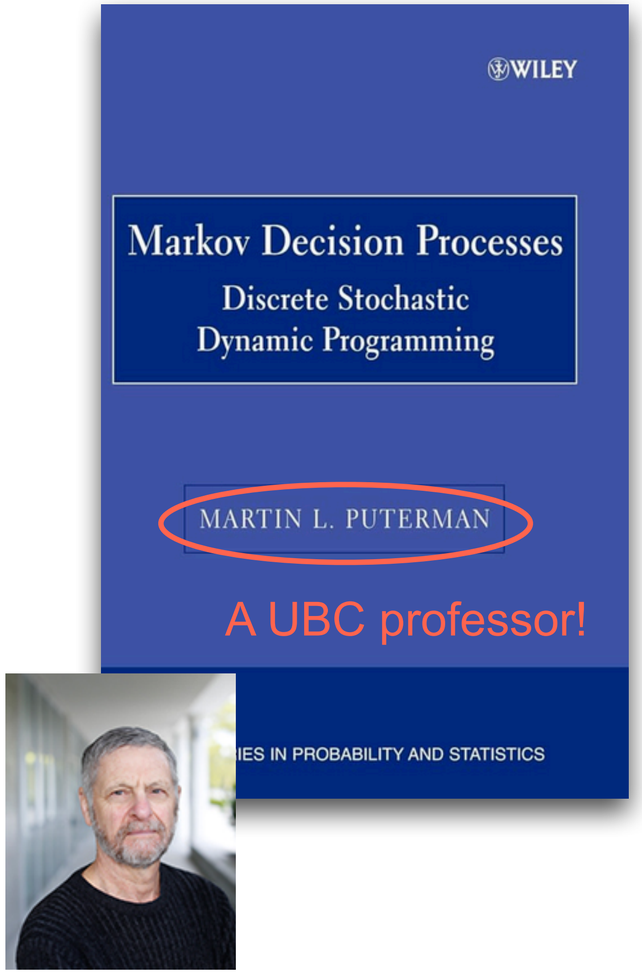
Today's lecture draws from chapters 2, 3, and 5

Modeling Sequential Decision-Making Problems



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Markov Decision Processes

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

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S - States

Set of all possible states from which the agent will make decisions.

Markov Decision Processes

A - Actions

Set of all possible actions the agent could take.

S - States

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$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

Markov Decision Processes

R- Reward function

$R(s, a) \in \mathbb{R}$ — A function defining the reward the agent receives for transitioning from state $s \in S$ to $s' \in S$ under action $a \in A$.

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$R(s,a,s')$

$R(s,a) = \mathbb{E}\left[\sum_{s'} T(s'|s,a) R(s,a,s')\right]$

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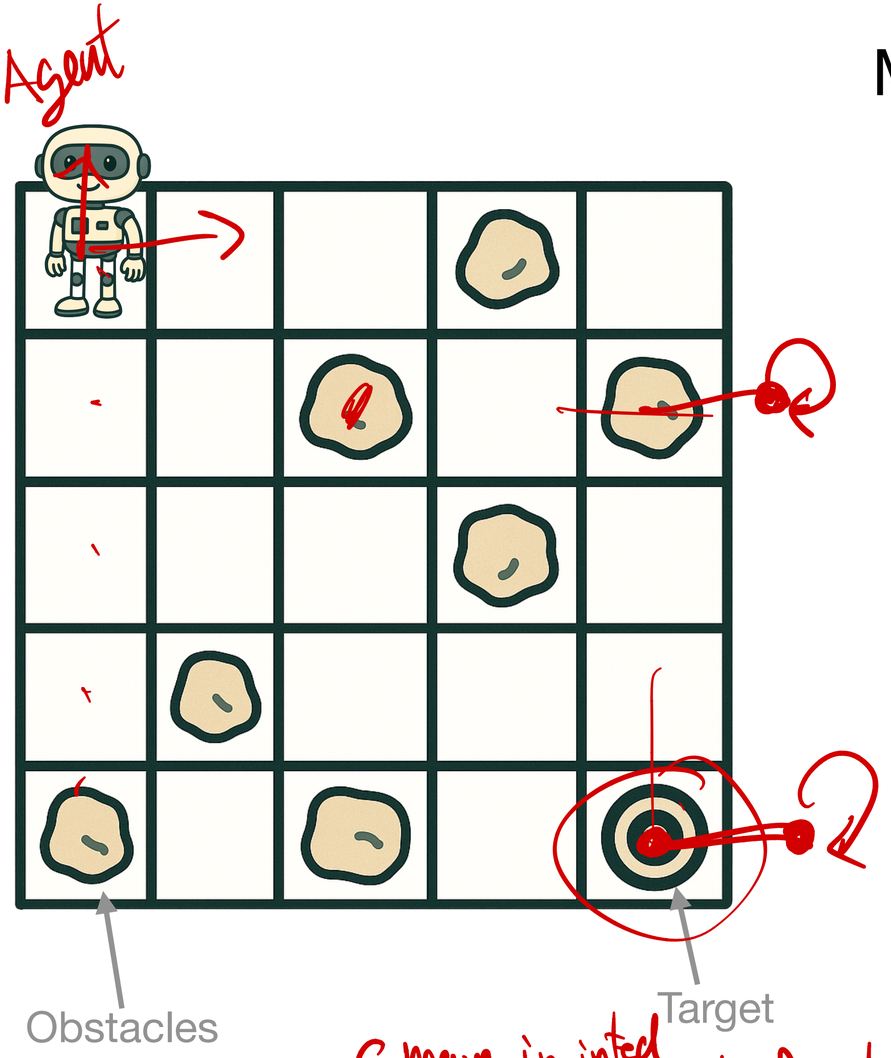
T - Transition function

$s' \sim T(\cdot \mid s, a)$ a stochastic transition function defining the probability of the agent transitioning to next state $s' \in S$, given that it executed action $a \in A$ from state $s \in S$.

MDP Examples: Gridworld Navigation



MDP Components:

State space S — All possible locations of the agent in the environment.

$S = \{(x, y) \mid x \in \{0,1,2,3,4\}, y \in \{0,1,2,3,4\}\}$

Action space A — Movement actions

$A = \{\text{left, right, up, down, diagonal, stay still}\}$

Transition function T — Move as planned/intended, so long as that action is valid, else, stay still. "Slip" with probability $p \in (0,1)$

$$T(s'|s,a) = \begin{cases} \text{move in inted direction} & \text{w.p. } 1-p \\ \text{move to random neighbor} & \text{w.p. } p \end{cases}$$

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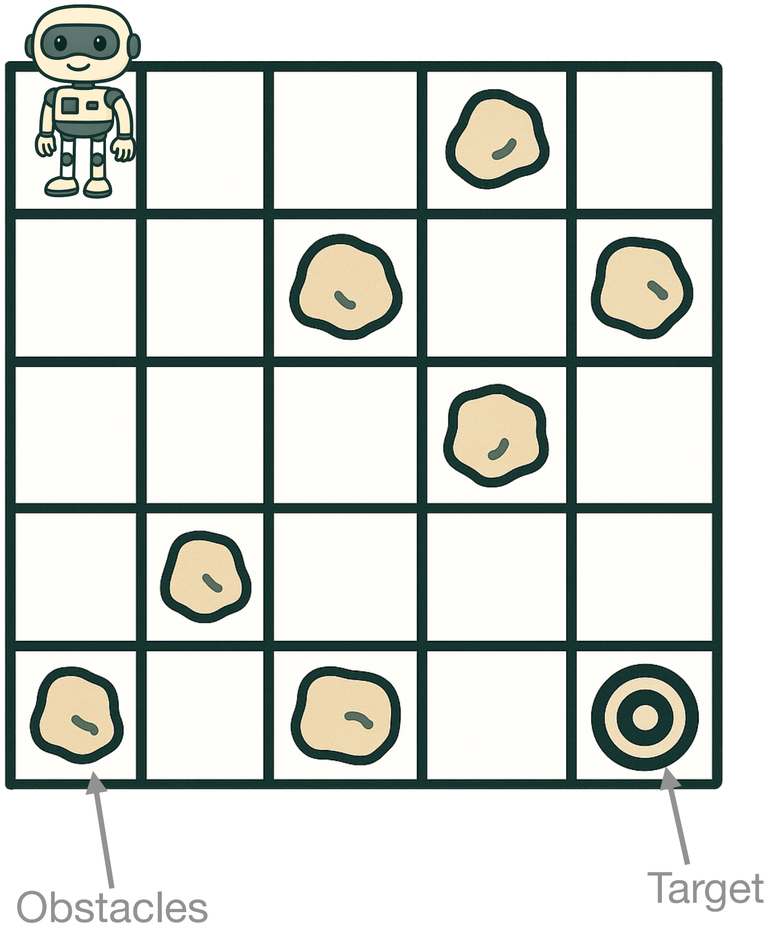


MDP Components:

Reward function R Reward the agent for reaching the goal, and penalize for hitting an obstacle.

$$R(s,a) = \begin{cases} +1 & \text{if } s = (4,4) \\ -1 & \text{if } s = \{(3,0), (2,1), \dots\} \end{cases}$$

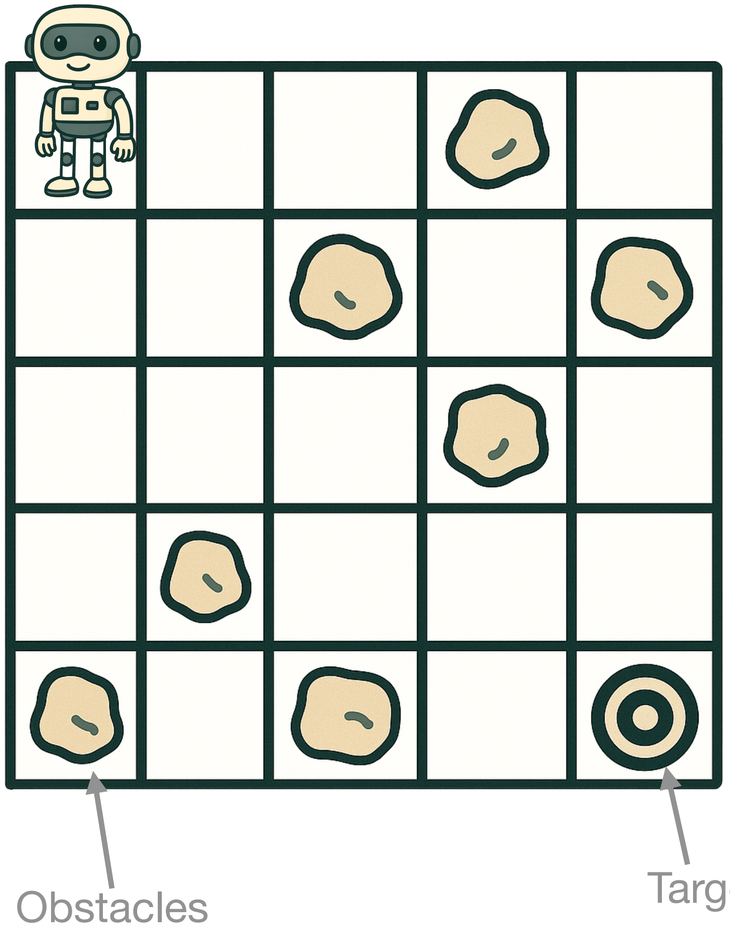
Initial distribution $\mu(s)$

We have options

$$\mu(s) = \begin{cases} 1 & \text{if } (0,0) = s_0 \\ 0 & \text{otherwise} \end{cases}$$

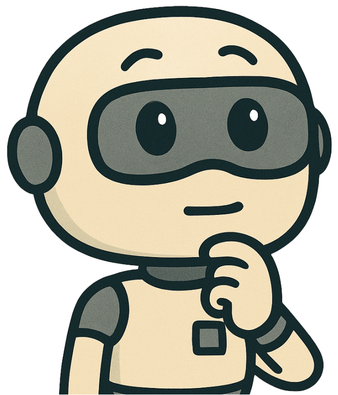
$\mu(s)$ = Uniform over states that can reach the goal.

MDP Examples: Gridworld Navigation



Why include stochastic transitions?
What could they model?

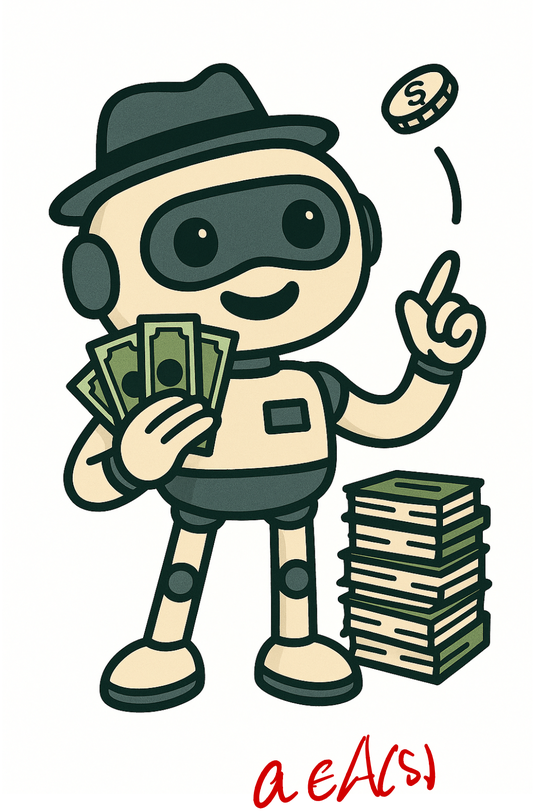
- Adding entropy to the problem that can help find the solution.
- When training it could help exploration
- Stochasticity can model uncertainty
  1. in environment transitions.
  2. Can also help make policies robust to observational uncertainty.



MDP Examples: Gambler's Ruin

$$T(s'|s,a) = \begin{cases} p & \text{if } s' = \min(\cdots) \\ 1-p & \text{if } s' = s-a \\ 0 & \text{otherwise} \end{cases}$$

- A gambler plays a game in which they make a bet of B dollars at every turn.
- They then have a probability p of winning $2B$ dollars, and a probability $1-p$ of losing the B dollars that they bet.
- They start with 50 dollars, and their goal is to cash out once they have have reached 100 dollars.



MDP Components:

State space S

The current amount of money the agent has

$S = \{0, 1, 2, \cdots, 100\}$

Action space A

They can bet an amount between 0 dollars and current state $s \in S$.

$A(s) = \{0, 1, 2, \cdots, s\}$

Transition function T

$$T(s'|s,a) = \begin{cases} s' = \min(100, s+a), & \text{w.p. } p \\ s' = s-a & \text{w.p. } 1-p \end{cases}$$

$a \in A(s)$

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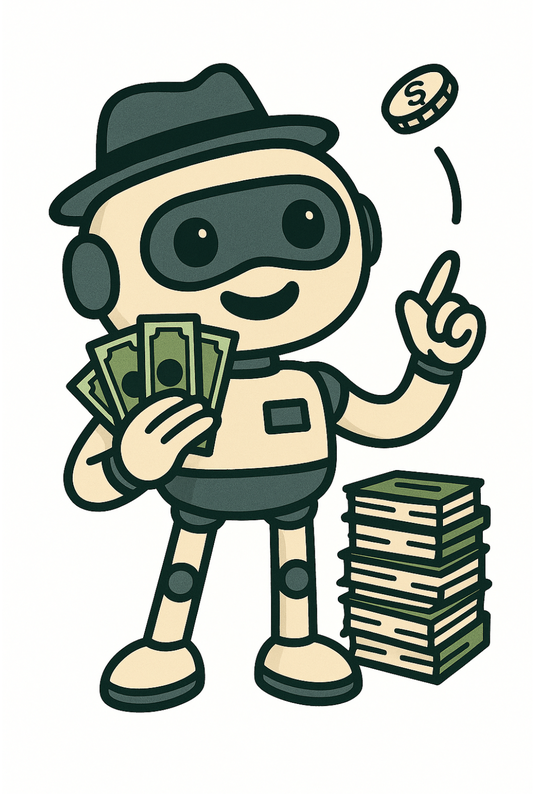


MDP Components:

Reward function R

Win 100 and cash out.

$$R(s,a,s') = \begin{cases} +1 & \text{if } s'=100 \\ 0 & \text{otherwise} \end{cases}$$

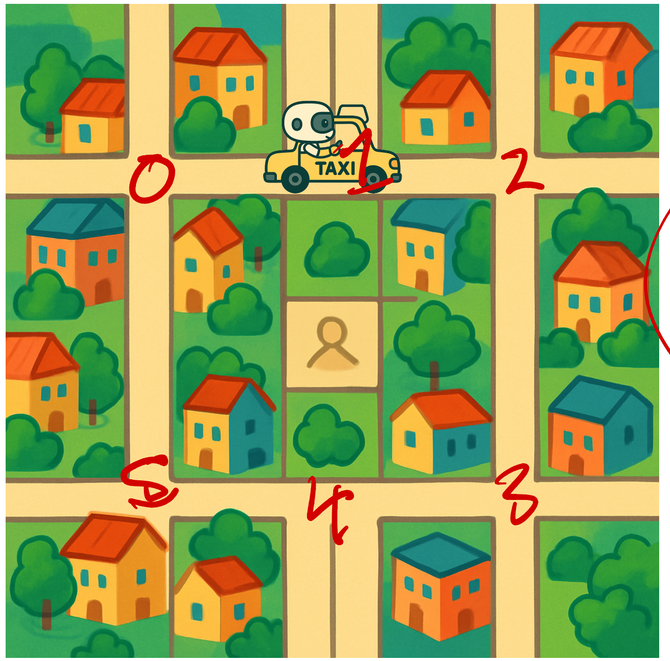
Initial distribution $\mu(s)$

$$\mu(s) = \begin{cases} 1 & \text{if } s_0 = 50 \\ 0 & \text{otherwise.} \end{cases}$$

Note: we <u>could</u> also define a distribution over various possible starting states.

MDP Examples: Autonomous Taxi Example

- An autonomous taxi needs to pickup and dropoff passengers in a city.
- Passengers randomly spawn at locations throughout the city, and each has a different dropoff location.
- The taxi can only carry at most 1 passenger at a time.
- The taxi gets $5 each time they deliver a passenger to the dropoff point.



MDP Components:

passenger flag

State space S

$S = \{0,1,2,3,4,5\} \times \{\text{index of which passenger}\}$
$\times \{\text{passenger destinations}\}$
$\times \{\text{passenger locations}\}$

Action space A

$A = \{\text{move up, left, down, right, pickup, dropoff}\}$

Transition function T

$T(s'|s,a)$

- Move from point to point
- Dropoff/pickup passengers according to action taken and the location of passengers.
- Spawn passengers randomly

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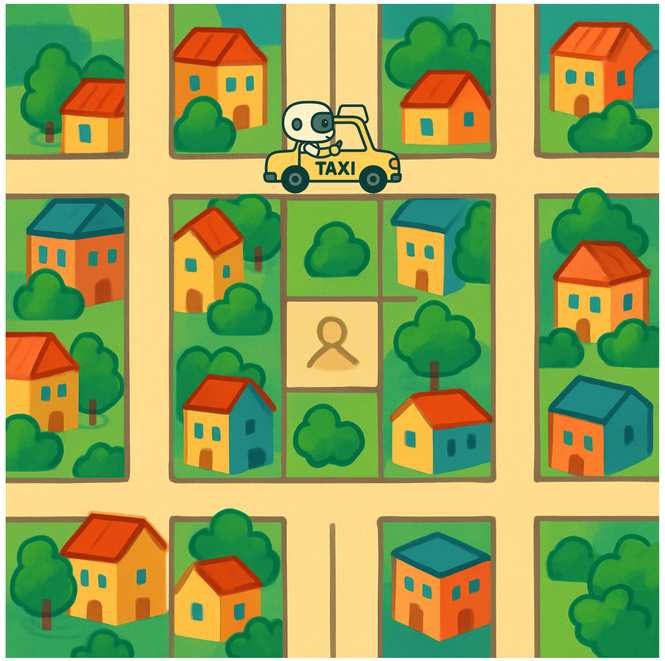

MDP Components:

Reward function R

$$R(s,a,s') = \begin{cases} +5 & \text{if dropoff in correct location} \\ 0, & \text{otherwise} \end{cases}$$

Initial distribution $\mu(s)$

- Random distribution for passenger locations and destinations
- Random distribution for taxi location.

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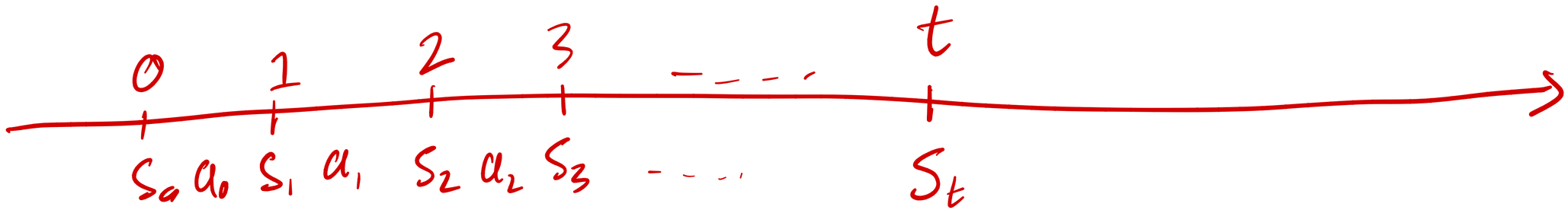

$$d_t(s_0, a_0, \ldots, s_t) \longrightarrow a \in A$$

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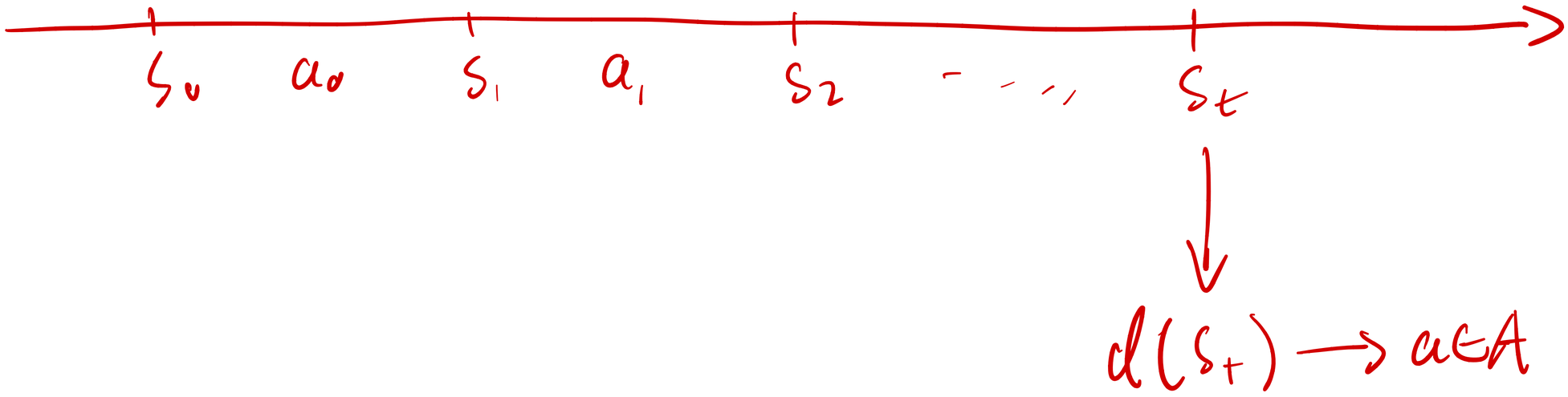

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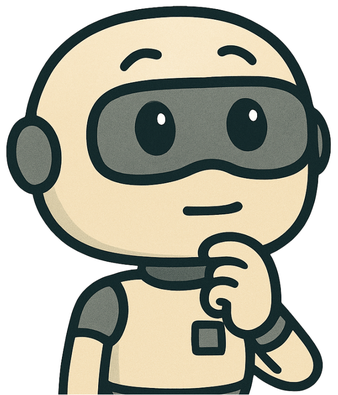
Decision rules can be:

	History-Dependent	**Markovian**
Deterministic	HD	MD
Random	HR	MR

When would we expect a decision rule to need history in order to make good decisions?

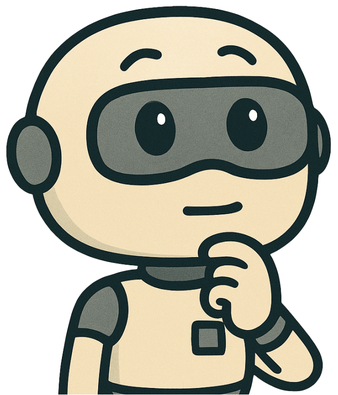
- when the current observation is not sufficient to fully encode the state information necessary for decision making.
  - e.g. • Chat bot needs history/context to understand prompt.
    - digital filters.

- If the state is "properly" defined, we won't need history.



When might a random decision rule be helpful over a deterministic one?

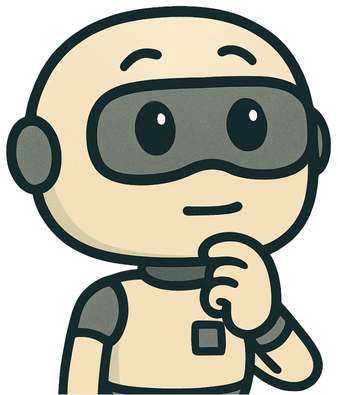
- In learning setting, we want to "explore". Random decision rule can help with this.
- In multiplayer/games, it can be necessary to have random decision rules to play optimally.
- Maybe it's easier to define/implement random decision rules.



More generally, why distinguish between different types of decision rules?

Each class of decision rules might have different algorithms for solving.

- we have to strike a balance between modeling fidelity (of the policy class) and computational burden of implementing the optimal policy.
  - How much info do we need to solve the problem optimally?
  - How much can we afford in terms of compute?



Policies and Policy Classes

A *policy* $\pi = d_0, d_1, \ldots$ is a sequence of decision rules that specifies which action the agent should take at **every** decision epoch.

policy π specifies

A policy belongs to a *policy class* $\Pi^{HD}, \Pi^{MD}, \Pi^{HR}, \Pi^{MR}$, if $\forall t,\ d_t \in D^{HD}, D^{MD}, D^{HR}, D^{MR}$, respectively.

e.g. $\pi \in \Pi^{MD}$ $d_0 \in D^{MD}, d_1 \in D^{MD}, \ldots, d_t \in D^{MD}$

A policy is *stationary* if the decision rule is constant in time, i.e. $d_t = d_{t'}$ for every t, t'.

i.e., only one decision rule d

In this course, we will often consider *stationary Markovian policies.*

Same decision rule at every timestep.
the decision rule is only a function of the current state.

What's the most general class of policies?

We considered many classes of policies.

$$\Pi^{SD} \subset \Pi^{SR} \subset \Pi^{MR} \subset \Pi^{HR}$$

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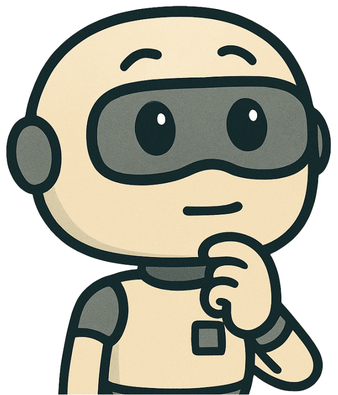
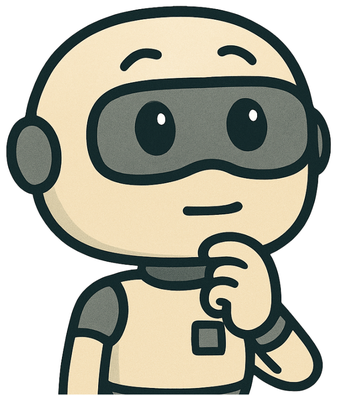
$$\Pi^{SD} \subset \Pi^{MD} \subset \Pi^{HD} \subset \Pi^{HR}$$



Why should we distinguish between stationary and non-stationary policies?

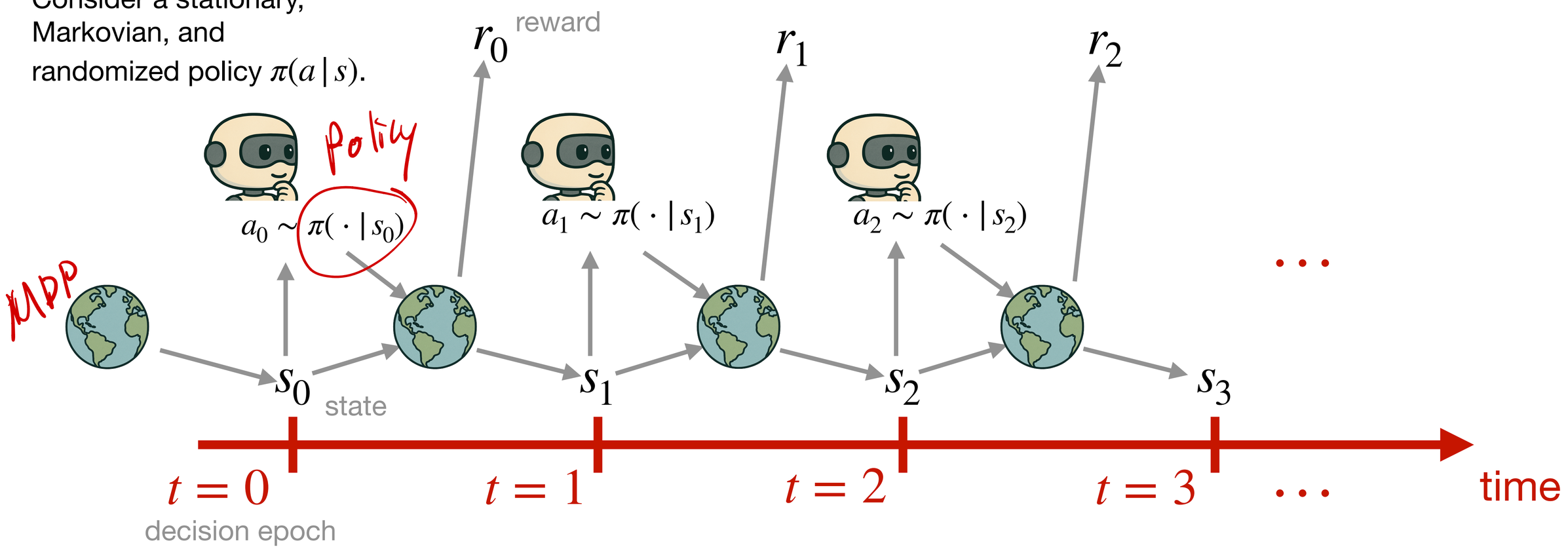
- Stationary policies are simpler, and necessary for infinite horizon problems.

- Non-stationary policies require lots of memory. Need to store separate decision rules for every timestep.



Executing a Policy in an MDP

Consider a stationary, Markovian, and randomized policy $\pi(a \mid s)$.



Executing a policy induces stochastic "trajectories" $\tau = s_0, a_0, r_0, s_1, a_1, r_1, \ldots$

Objective: choose $\pi(a \mid s)$ to maximize expectation of some function of trajectories.

$$\pi^* \in \text{argmax}_\pi \mathbb{E}[f(s_0, a_0, r_0, s_1, a_1, r_1, \ldots) \mid \pi]$$

Objective: Design “Performant” Policies w.r.t. Reward

A common optimization problem: Infinite-Horizon Total Discounted Reward

$$\pi^* \in \arg\max_{\pi \in \Pi^{HR}} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,|\, \pi, s_0 \sim \mu\right]$$

Intuition: Find a policy that maximizes the expected sum of discounted rewards.

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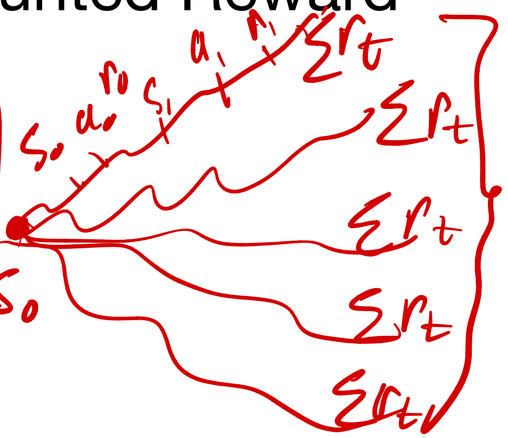
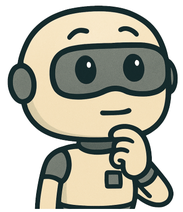


Intuition: Find a policy that maximizes the expected sum of discounted rewards.

What does this optimization problem actually mean?

- What distribution is the expectation taken over?
- Why is discounting necessary?
- When does such an optimal policy exist (particularly in a manageable policy class)?



Fixing a Policy in an MDP Induces Stochastic Processes

example with dice

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **finite time horizons**?

Ω Sample space, set of all possible outcomes $\omega \in \Omega$

e.g. for dice $\omega \in \{1,2,3,4,5,6\} = \Omega$

$\mathcal{F} \subseteq 2^\Omega$ "sigma algebra" of all possible "events".

↳ each event $A \in \mathcal{F}$, is a set of outcomes.

e.g. $A = \{2,4,6\}$ for "is the roll even"

$A = \{1\}$ for "did I role 1?"

$(\Omega, \mathcal{F}, \mathbb{P})$ — Probability space

$\mathbb{P} : \mathcal{F} \to [0,1]$ probability measure

e.g. $\mathbb{P}(A \text{ "roll even"}) = 1/2$

Random variable:

$X : \Omega \to \mathbb{R}$ is a measurable function w.r.t. probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **finite time horizons**?

Sample space $\Omega = S \times A \times S \times A \times \ldots \times S = \{S \times A\}^N \times S$

$$\omega = (s_0, a_0, s_1, a_1, \ldots, s_N) \in \Omega$$

Each outcome ω is a potential *trajectory* of length N.

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Probability measure

$$\begin{aligned}\mathbb{P}^\pi\left((s_0, a_0, s_1, a_1, \ldots, s_N)\right) &= \mu(s_0)\pi(a_0 \mid s_0)T(s_1 \mid s_0, a_0)\pi(a_1 \mid s_1)T(s_2 \mid s_1, a_1)\ldots \\ &= \mu(s_0)\prod_{t=0}^{N}\pi(a_t \mid s_t)T(s_{t+1} \mid s_t, a_t)\end{aligned}$$

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **infinite time horions**?

Fixing a Policy in an MDP Induces Stochastic Processes

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Events are so-called cylinder sets.

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Probability measure

Cylinder sets have probability mass. Infinite trajectories have measure zero.

$$\mathbb{P}^\pi\left((s_0, a_0, s_1, a_1, \dots, s_N)\right) = \mu(s_0)\pi(a_0 \mid s_0)T(s_1 \mid s_0, a_0)\pi(a_1 \mid s_1)T(s_2 \mid s_1, a_1)\dots$$

$$= \mu(s_0)\prod_{t=0}^{N} \pi(a_t \mid s_t)T(s_{t+1} \mid s_t, a_t)$$

Optimality Criterion Revisited

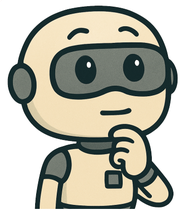
A common optimization problem: Infinite-Horizon Total Discounted Reward

$$\pi^* \in \arg\max_{\pi \in \Pi^{HR}} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,|\, \pi, s_0 \sim \mu\right]$$

Intuition: Find a policy that maximizes the expected sum of discounted rewards.

What does this optimization problem actually mean?

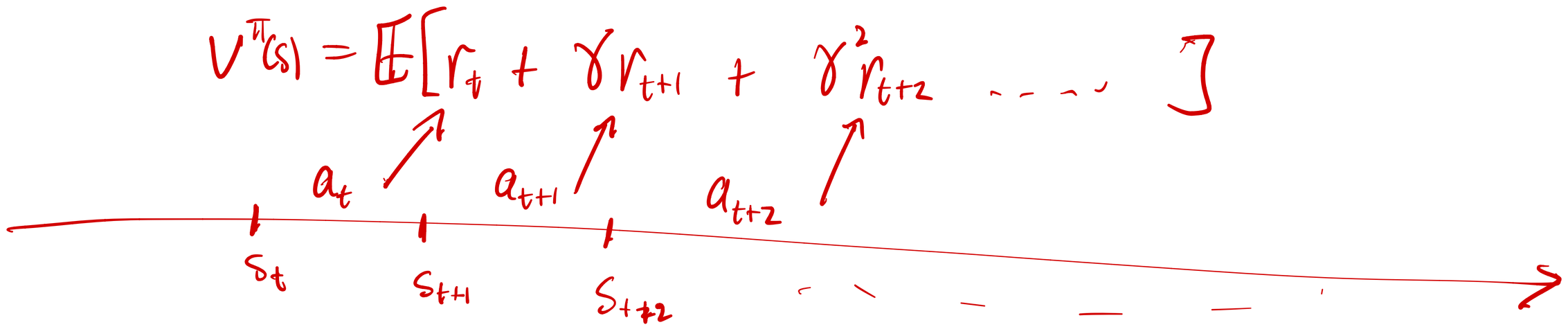
- What distribution is the expectation taken over?
- Why is discounting necessary?
- When does such an optimal policy exist (particularly in a manageable policy class)?



Defining the Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

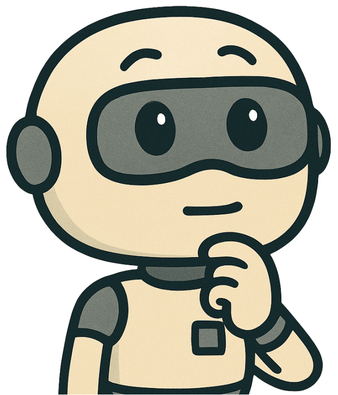
Intuition: How "valuable" is it for the agent to be in state s, if it follows policy π from that point onwards?



What Assumptions Did we Make When Defining the Value Function?

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

- We assumed that the infinite sum converges.
- We assumed $V^{\pi}(s)$ doesn't depend on the current time t.
  - ↳ This works for stationary policies and infinite time horizons and markovian states.



The Mathematical Importance of Discounting

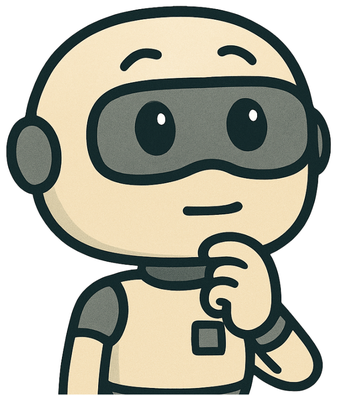
$$V^{\pi}(s) = \lim_{N \to \infty} \mathbb{E}\left[\sum_{t=0}^{N} \gamma^t r_t \mid \pi, s_0 = s\right]$$

When does the limit exist? Suppose $R(s,a)$ is bounded. $\exists M < \infty$ s.t. $|R(s,a)| \leq M$ $\forall s, a$

Consider $\lim_{N\to\infty} \mathbb{E}\left[\sum_{t=0}^{N} |\gamma^t r_t| \mid \pi, s_0\right] \leq \lim_{N\to\infty} \sum_{t=0}^{N} \gamma^t M$

$\gamma < 1$ $\lim_{N\to\infty} \sum_{t=0}^{N} \gamma^t M$ is a geometric series $\longrightarrow \frac{M}{1-\gamma}$

Series is absolutely convergent $\Rightarrow$ also convergent.



Why is this important? - Discounting is essential for the math of infinite horizon problems.

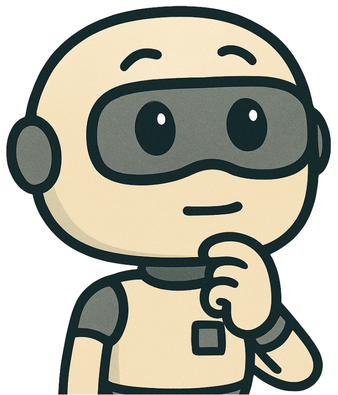
Otherwise values could be infinite and things get tricky.

When is/isn't the Value Function Stationary in Time?

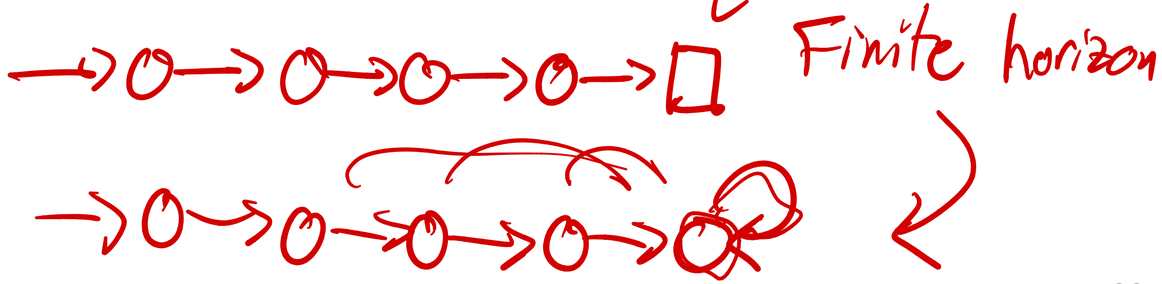
$$V^{\pi}(s) = \lim_{N \to \infty} \mathbb{E}\left[\sum_{t=0}^{N} \gamma^t r_t \mid \pi, s_0 = s\right]$$

- Stationary in time because MDP is stationary
  i.e. Dynamics, reward function, policy don't explicitly depend on time.

- It wouldn't necessarily be stationary if there was a finite time horizon.



Transforming finite horizon to infinite horizon.



Defining the Action-Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?

Defining the Action-Value Function

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Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?

$$Q^{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s, a_0 = a\right]$$

Intuition: How “valuable” is it for the agent to take action a from state s, if it follows policy π from that point onwards?

Defining the Action-Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?

$$V^{\pi}(s) = \sum_{a \in A} \pi(a \mid s) Q^{\pi}(s, a) \quad \updownarrow \quad Q^{\pi}(s, a) = R(s, a) + \gamma \sum_{s' \in S} T(s' \mid s, a) V^{\pi}(s')$$

$$Q^{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s, a_0 = a\right]$$

Intuition: How “valuable” is it for the agent to take action a from state s, if it follows policy π from that point onwards?

An Optimality Criterion: Infinite-Horizon Total Discounted Reward

A policy $\pi^*(\cdot \mid s)$ is considered optimal whenever

$$V^{\pi^*}(s) \geq V^{\pi}(s), \quad \forall s \in S, \forall \pi \in \Pi^{HR}$$

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The optimal value of an MDP is defined as:

$$V^*(s) := \sup_{\pi \in \Pi^{HR}} V^{\pi}(s)$$

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The optimal value of an MDP is defined as:

$$V^*(s) := \sup_{\pi \in \Pi^{HR}} V^{\pi}(s)$$

An optimal policy $\pi^* \in \Pi^{\{HR,MR,HD,MD\}}$ exists when:

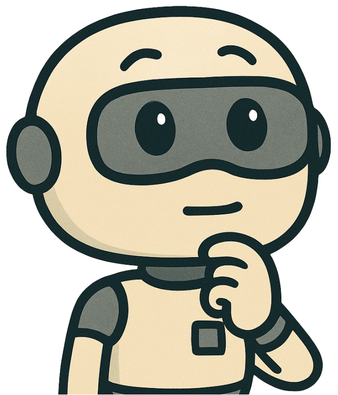
$$V^{\pi^*}(s) \geq V^*(s), \quad \forall s \in S$$

Optimality of Markovian Policies

What policy classes $\Pi^{HR}, \Pi^{HD}, \Pi^{MR}, \Pi^{MD}$ do we need to search to find an optimal policy?

- For any policy $\pi \in \Pi^{HR}$, for each s, there exists a policy $\pi' \in \Pi^{MR}$ for which $V^{\pi}(s) = V^{\pi'}(s)$
  Thm 5.5.3 in Puterman (for infinite horizon stationary MDPs)

- Also, there exists a policy $\pi \in \Pi^{MD}$ that is optimal (i.e. achieves at least as much value as any other policy).



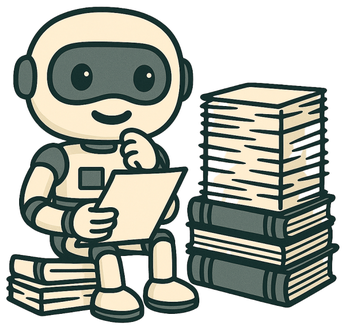
Other Optimality Criteria

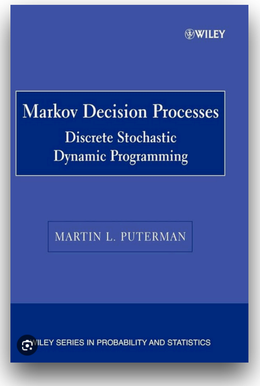
- So far, we've only presented one notion of optimality, assuming $0 \leq \gamma < 1$, and that our goal is to maximize the *sum* of future rewards.

- In this course, we will stick to discounted sums of reward over infinite horizons, due to the mathematical simplicity of this setting, and its dominance in the literature.

- Other notions of optimality in MDPs have been studied, and new ones can be defined (perhaps by you!).
  - E.g., Finite horizons, undiscounted rewards, average rewards, entropy-regularized value functions, risk-aware measures of value, etc.

- If you're interested in learning more, start by reading Puterman's book.





Chapters 4-10

Next Class: Solving for Optimal Policies